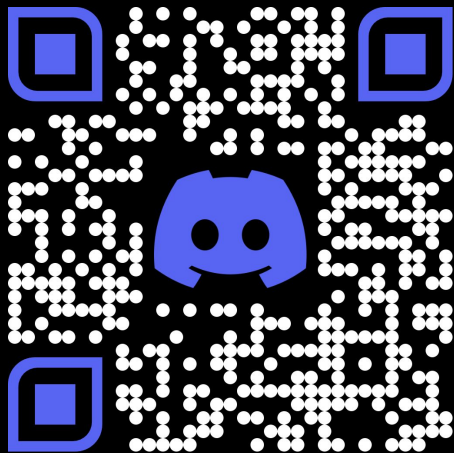


Capture the flag

DISCORD





Capture the flag



Capture the flag

- CTFs are team-based cybersecurity competitions, often involving real-world attacks
- You score points by *capturing the flag* of a given challenge, where the flag is a secret string hidden from you

Capture the flag

Challenges are divided into many categories:

- **Pwn**: Exploit binaries or services, bypass security measures, pop shells in random people's servers.
- **Crypto**: Study and break crypto protocols and implementations. Fun is guaranteed.
- **Web**: XSS attacks, CSRF vulnerabilities, backend misconfigurations. Everything that could help you get unauthorized control over a website
- **Reverse**: Break down and understand the logic of a binary. Think about old-school keygens or anything related to malware analysis.
- **Stegano, OSINT, Misc**: Join the team for more information :)

CTF is like football

- Effective age: 14-35 yrs old
- Play on weekends
- International mega tournaments
- Physically exhausting
- Mega teams like real madrid
- Buying/selling of players
- Doping



Doping

- Somehow still legal
- Good taste
- Sparkling
- Way too much sugar
- Goes directly to your brain
- No more sleep
- No more hunger
- Just focus on flagging



Skills tested

Skills tested

- Competitive programming

Gennady Korotkevich



Korotkevich in 2014

Born	September 23, 1994 (age 29) Gomel
Nationality	 Belarus
Other names	Codeforces handle : tourist
Education	ITMO University
Known for	Programming prodigy; highly ranked sport programmer from an early age
Awards	Codeforces peak rating 3979 (22 January 2022)

Skills tested

- Competitive programming
- Debugging



PS PlumbingSupply.com
How to Unclog a Toilet

Skills tested

- Competitive programming
- Debugging
- Assembly

```
section      .text
global      _start                      ;must be declared for linker (ld)

_start:                                           ;tell linker entry point

    mov     edx,len                      ;message length
    mov     ecx,msg                      ;message to write
    mov     ebx,1                        ;file descriptor (stdout)
    mov     eax,4                        ;system call number (sys_write)
    int     0x80                          ;call kernel

    mov     eax,1                        ;system call number (sys_exit)
    int     0x80                          ;call kernel

section      .data

msg         db  'Hello, world!',0xa        ;our dear string
len         equ $ - msg                   ;length of our dear string
```

Skills tested

- Competitive programming
- Debugging
- Assembly
- Complaining to the admins



Skills tested

- Competitive programming
- Debugging
- Assembly
- Complaining to the admins
- Command line magic
- ripgrep
- fd
- tmux
- bash for loops
- jq
- ssh
- man

Skills tested

- Competitive programming
- Debugging
- Assembly
- Complaining to the admins
- Command line magic
- Reading cursed crypto papers



Skills tested

- Competitive programming
- Debugging
- Assembly
- Complaining to the admins
- Command line magic
- Reading cursed crypto papers
- Guessing

It's 3am on the last night, the pwn players have all gathered around one screen, there is no leak, they're just trying to guess ASLR. 2006 [colorized]



Team rating

2022 2021 2020 2019 2018 2017 2016 2015 2014 2013
2012 2011

Place	Team	Country	Rating
👑 1	Project Sekai		228.454
2	r3kapig		169.606
3	u0K++		157.573
4	idek		140.009
5	WreckTheLine		126.752
6	C4T BuT S4D		118.427
7	Never Stop Exploiting		117.770
8	DiceGang		112.467
9	Chaitin Tech		109.760
10	perfect r00t		109.760

[Full rating](#) | [Rating formula](#)

Upcoming events 📅 🔊

Open [Finals](#)

Past events 🔊

With scoreboard [All](#)

🏠 Ugra CTF Quals 2022

March 01, 2022 07:00 UTC | On-line | [Weight voting in progress](#)

Place	Team	Country	Points
👑 1	s3qu3nc3		62.000
2	Project Sekai		45.988
3	FaKappa		37.234

[103 teams total](#) | [Tasks and writeups](#)

🏠 wtfctf 2022

Feb. 28, 2022 18:29 UTC | On-line | [Weight voting in progress](#)

Place	Team	Country	Points
👑 1	Project Sekai		38.420
2	1nf1n1ty		27.007
3	Flag Poisoning		22.772

[246 teams total](#) | [Tasks and writeups](#)

🏠 TSJ CTF 2022

Feb. 28, 2022 05:00 UTC | On-line | [Weight voting in progress](#)

The teams

polygl0ts



<https://polygl0ts.ch>

- It's us!
- Academic EPFL team

The teams

polygl0ts



<https://polygl0ts.ch>

- It's us!
- Academic EPFL team

flagbot



<https://flagbot.ch>

- “The grass is always greener on the other side”
- Academic ETH team

The teams

polygl0ts



<https://polygl0ts.ch>

- It's us!
- Academic EPFL team

flagbot



<https://flagbot.ch>

- “The grass is always greener on the other side”
- Academic ETH team

organizers



<https://org.anize.rs/>

- Born from a meme
- very strong
- polygl0ts + flagbot + cr0wn + some small other teams

The teams

polygl0ts



Lausanne Sport

- Play for fun
- Fair play
- Learn together
- Cute small local team

flagbot



F.C. Zurich

- Sister team
- Currently in a lesbian relationship with us
- Pretend to compete but it's just foreplay

organizers



Real Madrid

- Play to win
- Top tier international finals
- Tryhard behaviour (doping, swearing, becoming furry, etc.)
- Cursed team, constant drama

Organizers very strong

#1 in the world 4 years ago

Always participating in DEFCON
FINALS

2022	2021	2020	2019	2018	2017	2016	2015	2014	2013
2012	2011								
Place	Team	Country					Rating		
👑 1	perfect blue						1409.969		
2	organizers						1173.314		
3	Super Guesser						1036.950		
4	Balsn						1010.773		
5	DiceGang						951.334		
6	More Smoked Leet Chicken						901.412		
7	Bushwhackers						819.242		
8	Plaid Parliament of Pwning						808.852		
9	Never Stop Exploiting						794.366		
10	Nu1L						776.762		
Full rating Rating formula									

Always participating in DEFCON FINALS

2023	2022	2021	2020	2019	2018	2017	2016	2015	2014
2013	2012	2011							

Place	Team	Country	Rating
👑 1	organizers		1511.323
2	perfect r00t		1283.824
3	Never Stop Exploiting		1025.931
4	C4T BuT S4D		1013.462
5	Water Paddler		942.923
6	Nu1L		903.930
7	justCatTheFish		887.498
8	Bushwhackers		817.228
9	DiceGang		756.228
10	kalmarunionen		737.250

Full rating | [Rating formula](#)

Organizers very strong

#1 in the world two years ago

Always participating in DEFCON
FINALS

Took part in many top-tier CTFs:
- ProCTF (Abu Dhabi)

2023	2022	2021	2020	2019	2018	2017	2016	2015	2014
2013	2012	2011							
Place	Team	Country	Rating						
👑 1	Blue Water		1272.461						
2	kalmarunionen		1030.562						
3	C4T BuT S4D		1010.123						
4	justCatTheFish		887.259						
5	r3kapig		822.413						
6	SKSD		761.927						
7	organizers		708.467						
8	thehackerscrew		698.748						
9	Project Sekai		684.336						
10	WreckTheLine		677.514						

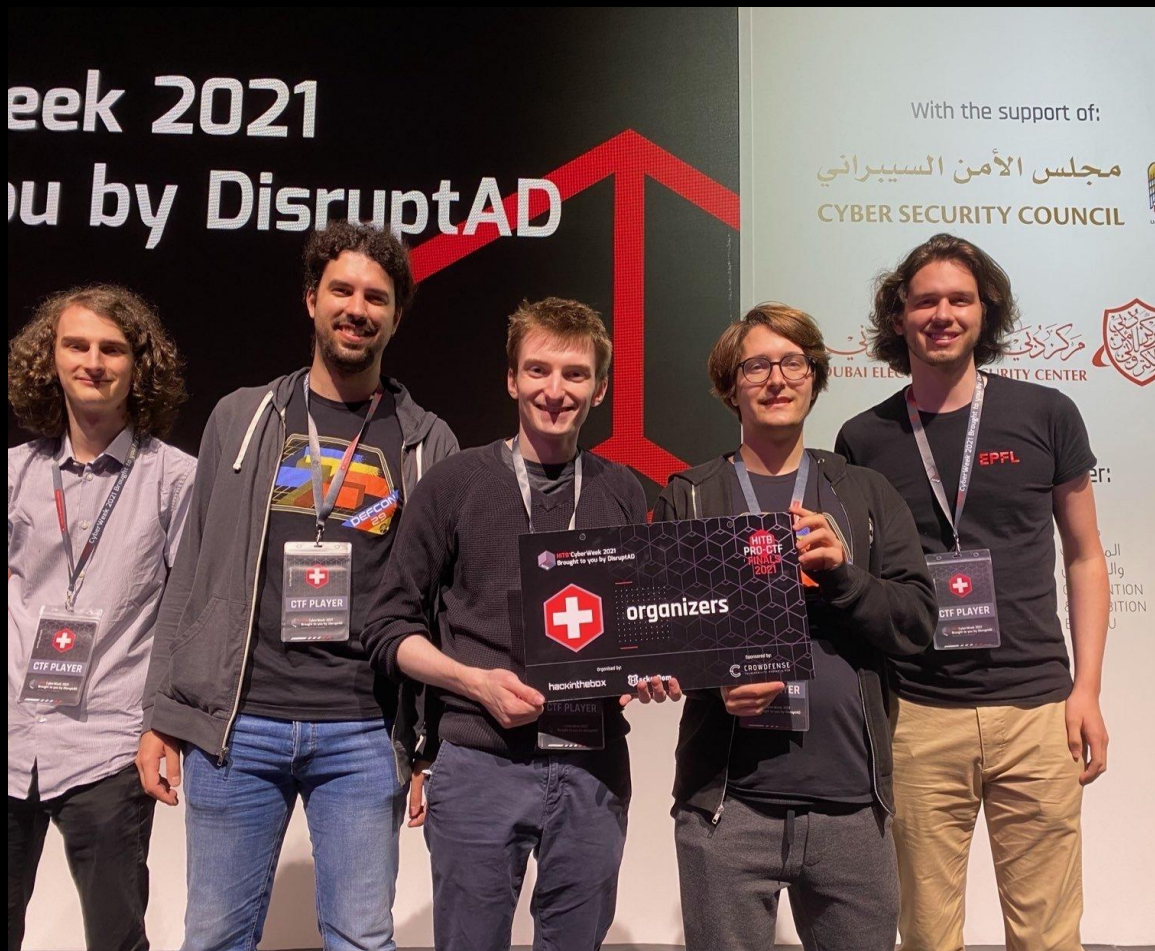
[Full rating](#) | [Rating formula](#)

Organizers very strong

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- ProCTF (Abu Dhabi)



Organizers very strong

#1 in the world two years ago

Always participating in DEFCON
FINALS

Took part in many top-tier CTFs:

- ProCTF (Abu Dhabi)
- DEFCON FINALS (Las Vegas)



Organizers very strong

#1 in the world two years ago

Always participating in DEFCON
FINALS

Took part in many top-tier CTFs:

- ProCTF (Abu Dhabi)
- DEFCON FINALS (Las Vegas)
- Google CTF (London, Tokyo)



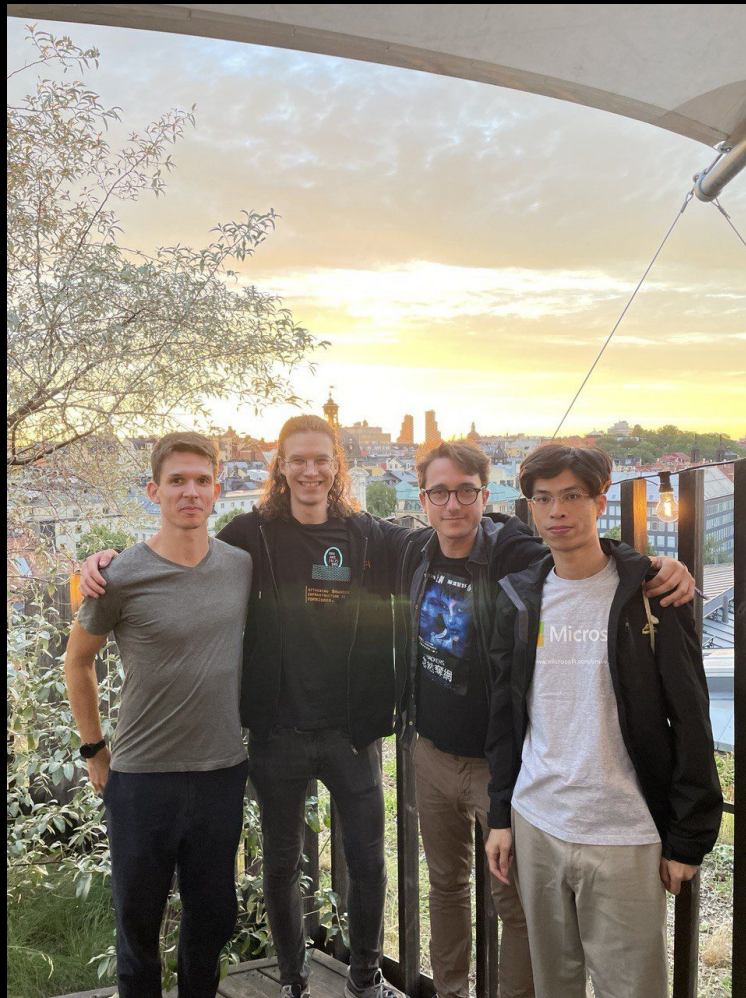
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- ProCTF (Abu Dhabi)
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- Google CTF (London, Tokyo)
- Midnight Sun (Stockholm)



Organizers very strong

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- XCTF (Chengdu, China)



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- Midnight Sun (Stockholm)
- XCTF (Chengdu, China)
- MOLECON (Torino)
- SECCON (Seoul)



Organizers very strong

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- Google CTF (London, Tokyo)
- Midnight Sun (Stockholm)
- XCTF (Chengdu, China)
- MOLECON (Torino)
- SECCON (Seoul)
- HXP CTF (Hamburg)
- HITCON
- Midnight Sun CTF (Stockholm)
- HackASat (Las Vegas)
- Insomnihack (Geneva)



How to git gud

- CTFs are hard, very involved
- Easy to get discouraged

How to get better:

- Play with people of your level
- Play with people above your level
- Read writeups

Writeups

> Very often it will happen that a chall is too hard for you.

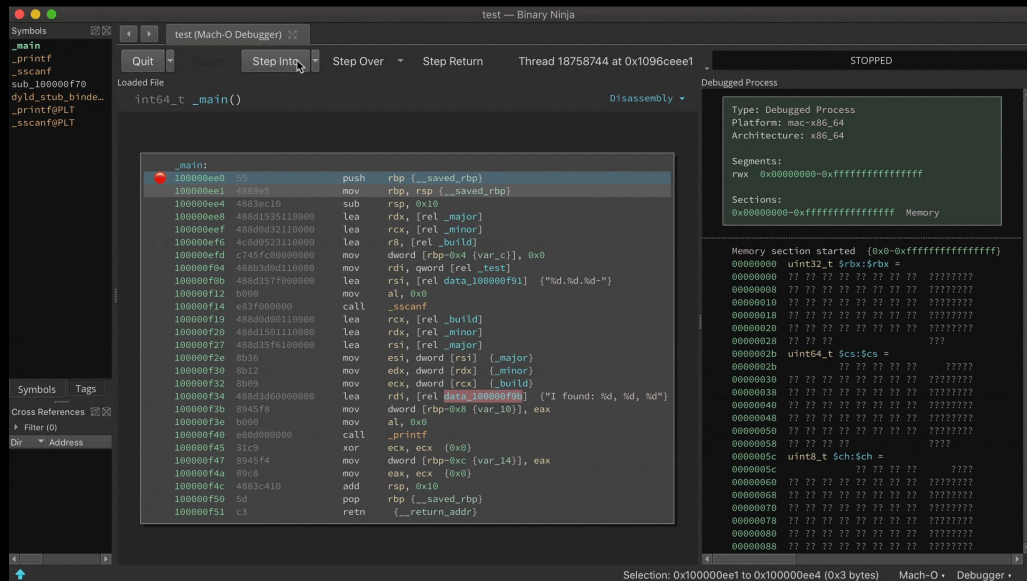
Take time after the CTF and read a writeup!

> Sometimes you'll be able to solve a challenge.

Write a writeup!!!

Tools

- ghidra/IDA/binja



Tools

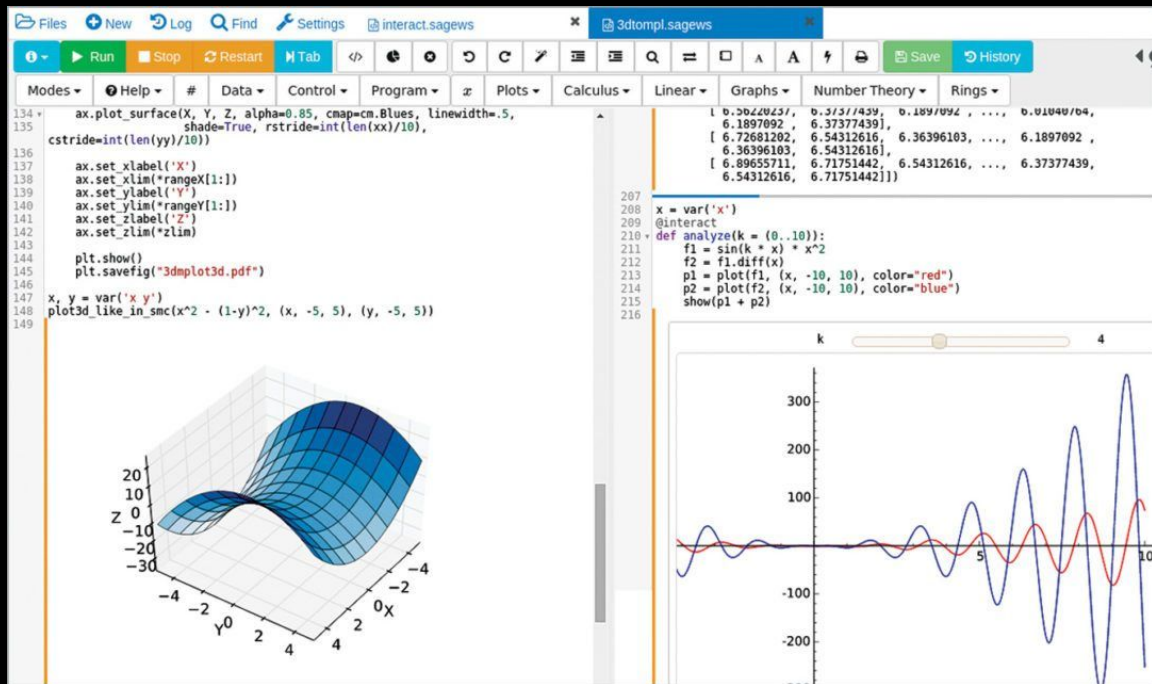
- ghidra/IDA/binja
- pwntools



```
from pwn import *  
r = remote("127.22.12.3", 8000)  
r.sendline(asm(shellcraft.sh()))  
r.interactive()
```

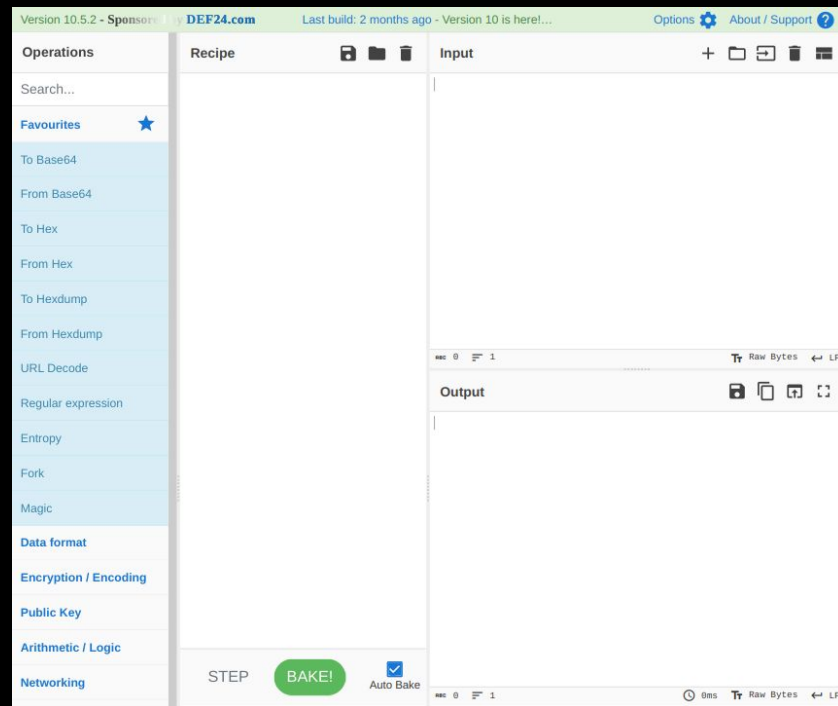
Tools

- ghidra/IDA/binja
- pwntools
- sage



Tools

- ghidra/IDA/binja
- pwntools
- sage
- cyberchef



Tools

- ghidra/IDA/binja
- pwntools
- sage
- cyberchef
- Z3

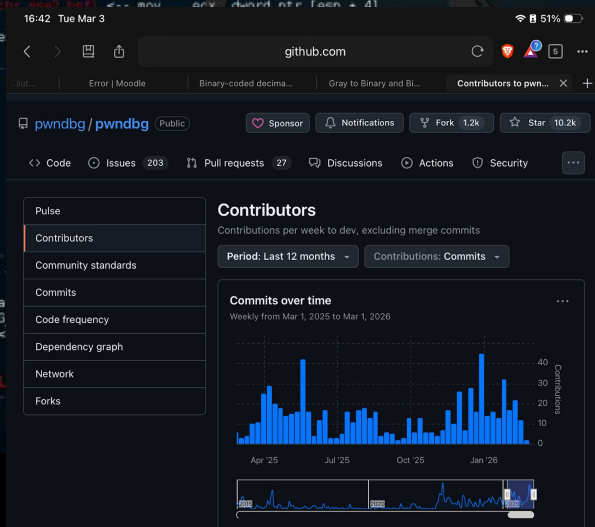


Tools

- ghidra/IDA/binja
- pwntools
- sage
- cyberchef
- Z3
- gdb

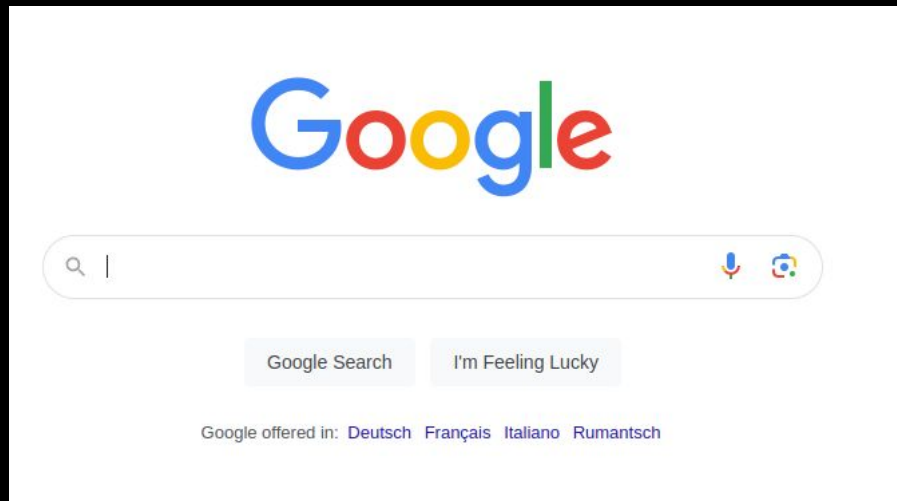
```
pwndbg> b *main
Breakpoint 1 at 0x8048deb
pwndbg> r
Starting program: /data/crackme

Breakpoint 1, 0x8048deb in main ()
LEGEND: STACK | HEAP | CODE | DATA | RWX | RODATA
[-----]
*EAX 0xffffd4a4 --> 0x80481d0 (_init) <-- push ebx
*EBX 0x80481d0 (_init) <-- push ebx
*ECX 0x3ea8d01
*EDX 0xffffd4a4 --> 0x80481d0 (_init) <-- push ebx
EDI 0x0
*ESI 0x071a00c (_GLOBAL_OFFSET_TABLE_+12) --> 0x0076c10 (_rawmain@plt) <-- mov ebx, dword ptr [esi + 4]
EBP 0x0
*ESP 0xffffd48c --> 0x80492e7 (_libc_start_main+487) <-- add esp, 4
*EIP 0x8048deb (main) <-- lea ecx, [esp + 4]
[-----]
=> 0x8048deb <main>    lea    ecx, [esp + 4]
0x8048def <main+4>    and    esp, 0xffffffff
0x8048df2 <main+7>    push   dword ptr [ecx - 4]
0x8048df5 <main+10>   push   ebp
0x8048df6 <main+11>   mov    ebp, esp
0x8048df8 <main+13>   push   esi
0x8048df9 <main+14>   push   ebx
0x8048dfa <main+15>   push   ecx
0x8048dfb <main+16>   sub    esp, 0x4c
0x8048dfe <main+19>   mov    ebx, ecx
0x8048e00 <main+21>   cmp    dword ptr [ebx], 1
[-----]
00:0000 esp    0xffffd48c --> 0x80492e7 (_libc_start_main+487)
01:0004    0xffffd490 <-- 0x1
02:0008    0xffffd494 --> 0xffffd504 --> 0xffffd67f <-- '/da
03:000c    0xffffd498 --> 0xffffd50c --> 0xffffd58d <-- 'XDG
04:0010    0xffffd49c --> 0xffffd4a4 --> 0x80481d0 (_init) <
05:0014    0xffffd4a0 <-- 0x0
06:0018 eax edx 0xffffd4a4 --> 0x80481d0 (_init) <-- push ebx
07:001c    0xffffd4a8 --> 0x071a00c (_GLOBAL_OFFSET_TABLE_+1
[-----]
> f 0 8048deb main
f 1 80492e7 _libc_start_main+487
Breakpoint *main
```



Tools

- ghidra/IDA/binja
- pwntools
- sage
- cyberchef
- Z3
- gdb
- google



LakeCTF

CTF we are hosting

November 27-28 quals

Finals in May (Balelec
afterparty!)

call for challenges!



Starts in: 37d 19h 39m 35s

LakeCTF is a jeopardy-style security hacking contest organized by the [polygl0ts](#) CTF team (plus some friends!) and the [EPFL School of Computer and Communication Sciences \(IC\)](#).

You must first compete in an online qualification round, and then physically compete in the finals in Lausanne. The top teams will have the unique chance to meet the top members of the EPFL research community.

We wish you the best of luck !

> Quals schedule

Qualification start	Sat 4th of November, 18:00 UTC
Qualification end	Sun 5th of November, 18:00 UTC

> Finals schedule

Fri 27th of April 2024	Arrival Tour of Lausanne
Sat 28th of April 2024	9:00 CTF Finals Start

Recap

WRITEUPS

0 PUBLISHED

ALL



Bronze

Silver

Gold

◆ Nine

REV

REV

Secure Flag Checker

NO WRITEUPS YET

Nothing published yet.

0 WRITEUPS

READ +

REV

Random means secure

NO WRITEUPS YET

Nothing published yet.

0 WRITEUPS

READ +

REV

Dark Souls Trusty Friend

NO WRITEUPS YET

Nothing published yet.

0 WRITEUPS

READ +

PWN

PWN

Are you admin?

NO WRITEUPS YET

Nothing published yet.

0 WRITEUPS

READ +

PWN

GD bee?

NO WRITEUPS YET

Nothing published yet.

0 WRITEUPS

READ +

PWN

Step Up

NO WRITEUPS YET

Nothing published yet.

0 WRITEUPS

READ +

PWN

Days Past

NO WRITEUPS YET

Recap

- Website: `polygl0ts.ch`
- Discord! (link in website)
- CTFs usually on weekends
- Weekly tutorials on Fri 17:00, for newcomers

